Failure Analysis Report

FAIL-0095 — Network Failure

1. Failure Summary

Failure ID	FAIL-0095
Failure Type	Network Failure
Severity	Major
Component	Network Stack
Application	APP-012 — SecureAuthentication
Build Version	4.3.3
Timestamp	2024-01-24 09:59:12
Environment	ENV-006 — Cloud Test Env
OS Version	Ubuntu 22.04
Network	Ethernet
Resolution	Resolved

2. Test Case Reference

Test Case ID	TC-0195
Title	Test that payment processing correctly handles invalid input when the target dev...
Execution ID	EXEC-00416
Test Run	RUN-006 — Regression Cycle 6 — Build 5.2.1
Requirement	REQ-0076 — Recovery — Bluetooth Reconnection

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete device pairing within the defined SLA.
Observed Behaviour	The system exhibited network failure when the input data contains Unicode special characters.

4. Error Details

Error Message:

```
SocketException: Connection reset by peer on interface eth0.
```

Stack Trace / Log Excerpt:

```
com.samsung.wallet.SensorManager.execute(line 159)
```

Log Reference	LOG-49686

5. Executed Steps at Time of Failure

1	Capture a screenshot and attach as evidence.	The audit log records the event with correct metadata.

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2	Re-connect the phone and confirm auto-recovery.	No crash or ANR is observed.
3	Capture a screenshot and attach as evidence.	A confirmation notification appears on the paired device.
4	Capture a screenshot and attach as evidence.	The device reconnects automatically within 10 seconds.
5	Check server logs for event code 'EVT-2223'.	The UI transitions to the next screen within 3 seconds.

6. Root Cause Analysis

Cache eviction policy evicts frequently-used configuration entries under memory pressure.

7. Linked Defect

Defect ID	BUG-0416
Title	Network Failure in Network Stack: SocketException: Connection reset by peer on i...
Severity	Major
Priority	Medium
Status	IN_PROGRESS
Root Cause	Cache eviction policy evicts frequently-used configuration entries under memory ...
Resolution	—
Assigned To	ENG-090

8. Historical Similar Failures

HIST-0044	FAIL-0045	Network Failure	5.2.9	2023-11-11	10	Wont Fix
HIST-0094	FAIL-0095	Network Failure	6.2.4	2022-05-10	11	Open
HIST-0158	FAIL-0159	Network Failure	3.4.9	2021-04-04	11	Resolved

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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